

EXISTENCE OF MINIMAL MODELS FOR VARIETIES OF LOG GENERAL TYPE II: PL-FLIPS

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ABSTRACT. Assuming finite generation in dimension $n - 1$, we prove that pl-flips exist in dimension n .

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1. INTRODUCTION

Recall the following well known conjecture.

Conjecture 1.1. *Let (X, Δ) be a kawamata log terminal pair of dimension n and let $f: X \rightarrow Z$ be a projective morphism to a normal affine variety $Z = \text{Spec } A$.*

If $k(K_X + \Delta)$ is Cartier and big, where k is a positive integer, then $R(X, k(K_X + \Delta))$ is a finitely generated A -algebra.

The purpose of this paper is to prove:

Theorem 1.2. *Assume $(1.1)_{n-1}$.*

Then pl-flips exist in dimension n .

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We remark that together with the results of [2], this implies that flips exist in all dimensions and that (1.1) holds in all dimensions. The results of this paper originally appeared in [3] which was submitted to the Journal of the American Mathematical Society. After the results of [2], we felt that it would be more appropriate to resubmit the paper [3] (under a different title and after a substantial revision) together with [2] as part of a series of two papers that give a complete proof of the existence of relative log terminal and log canonical models (and hence also of flips) for varieties of log general type of any dimension.

The main ideas used in this paper have their origins in the work of Shokurov on the existence of flips [9] together with the use of the extension theorem of [4] which in turn was inspired by work of Kawamata, Siu and Tsuji (cf. [6], [10] and [13]). In this paper, however we do not make use of the concept of “asymptotic saturation” introduced by Shokurov, and in fact we prove a more general result for restricted algebras which does not require the relative weak log-Fano condition (see also [1]).

Further treatments of the results of this paper may be found in [1] and [5] (which follows Shokurov’s approach more explicitly).

We now turn to a more detailed description of the results and techniques used in this paper. Recall the following:

Definition 1.3. *Let (X, Δ) be a purely log terminal pair and $f: X \rightarrow Z$ be a projective morphism of normal varieties. Then f is a **pl-flipping contraction** if Δ is a $\mathbb{Q}$ -divisor and*

- (1) *f is a small birational morphism of relative Picard number one,*
- (2) *$-(K_X + \Delta)$ is f -ample,*
- (3) *X is $\mathbb{Q}$ -factorial,*
- (4) *$S = \lfloor \Delta \rfloor$ is irreducible and $-S$ is f -ample.*

*The **flip** of a pl-flipping contraction $f: X \rightarrow Z$ is another small birational projective morphism $g: Y \rightarrow Z$ of relative Picard number one, such that $K_Y + \Gamma$ is g -ample.*

The flip g is unique, if it exists at all, and it is given by

$$Y = \text{Proj}_Z \mathfrak{R} \quad \text{where} \quad \mathfrak{R} = \bigoplus_{m \in \mathbb{N}} f_* \mathcal{O}_X(mk(K_X + \Delta)),$$

and k is any positive integer such that $k(K_X + \Delta)$ is integral. Therefore, in order to prove the existence of pl-flips, it suffices to show that $\mathfrak{R}$ is a finitely generated $\mathcal{O}_Z$ -algebra. Since this problem is local over Z , we may assume that $Z = \text{Spec } A$ is affine and it suffices to prove that

$$R(X, k(K_X + \Delta)) = \bigoplus_{m \in \mathbb{N}} H^0(X, \mathcal{O}_X(mk(K_X + \Delta))),$$

is a finitely generated A -algebra. It is then natural to consider the restricted algebra

$$R_S(X, k(K_X + \Delta)) = \text{Im}(R(X, k(K_X + \Delta)) \longrightarrow R(S, k(K_S + \Theta))),$$

whose graded pieces correspond to the images of the restriction homomorphisms

$$H^0(X, \mathcal{O}_X(mk(K_X + \Delta))) \longrightarrow H^0(S, \mathcal{O}_S(mk(K_S + \Theta))),$$

where Θ is defined by the adjunction formula

$$(K_X + \Delta)|_S = K_S + \Theta,$$

and $k(K_X + \Delta)$ is Cartier. Shokurov has shown, cf. (3.2), that the algebra $R(X, k(K_X + \Delta))$ is finitely generated iff the restricted algebra is finitely generated.

Now, if the natural inclusion $R_S(X, k(K_X + \Delta)) \subset R(S, k(K_S + \Theta))$ were an isomorphism, then (1.2) would follow easily. In fact the pair (S, Θ) is kawamata log terminal, and since $f|_S$ is birational, $K_S + \Theta$ is automatically big. Since $\dim S = \dim X - 1 = n - 1$, $R(S, k(K_S + \Theta))$ is finitely generated by assumption and (3.2) implies $\mathfrak{R}$ is also finitely generated.

Unluckily this is too much to hope for. However it does suggest that one should concentrate on the problem of lifting sections and the main focus of this paper is to prove two extension results, (4.1) and (6.1). In fact (1.2) is a straightforward consequence of these results.

Let us start with an example where we cannot lift sections. Let X be the blow up of $\mathbb{P}^2$ at a point p , with exceptional divisor E . Let S be the strict tranform of a line through p .

We explain the second extension result (6.1) first.

§5 and §7. The conclusion, cf. (6.1), is that we can extend sections provided the stable base locus of $K_X + \Delta$ does not contain any canonical centres of (S, Θ) .

Unfortunately this lifting result is not quite enough to prove finite generation directly.

We use this result in §7

2. NOTATION AND CONVENTIONS

We work over the field of complex numbers $\mathbb{C}$. Let X be a normal variety. Let $K \in \{\mathbb{Q}, \mathbb{R}\}$. A K -divisor D is a K -linear combination of prime divisors. Note that the group of Weil divisors with coefficients in K forms a vector space, with a canonical basis given by the prime

divisors. If $C = \sum c_i B_i$ and $D = \sum d_i B_i$, then we will denote by

$$\begin{aligned} \|C\| &= \max_i c_i & \lceil C \rceil &= \sum_i \lceil c_i \rceil B_i \\ \{C\} &= \lceil C \rceil - C & C \wedge D &= \sum_i \min\{c_i, d_i\} B_i. \end{aligned}$$

Two $\mathbb{R}$ -divisors C and D are *K-linearly equivalent* $C \sim_K D$ if their difference is an K -linear combination of principal divisors. The *K linear system* of a K -divisor D is

$$|D|_K = \{C \geq 0 \mid C \sim_K D\}.$$

If D is a $\mathbb{Q}$ -divisor, then the *stable base locus* of D is the intersection of the support of the elements of the rational linear system $|D|_{\mathbb{Q}}$. The *stable fixed divisor* is the divisorial support of the stable base locus. If D is big, then the *augmented stable base locus* of D denoted by $\mathbf{B}_+(D)$ is given by the stable base locus of $D - \epsilon A$ for some ample divisor A and any rational number $0 < \epsilon \ll 1$. If D is an integral divisor, $\text{Fix}(D)$ denotes the fixed part of D so that $|D| = |D - \text{Fix}(D)| + \text{Fix}(D)$ where the base locus of $|D - \text{Fix}(D)|$ contains no divisors.

A *K-Cartier divisor* D is a K -linear combination of Cartier divisors. A K -Cartier divisor D is *nef* if $D \cdot \Sigma \geq 0$ for any curve $\Sigma \subset X$. A K -Cartier divisor D is *ample* if it is K -linearly equivalent to a positive linear combination of ample divisors (in the usual sense). A K -divisor D is *effective* if it is a positive real linear combination of prime divisors. A K -Cartier divisor D is *big* if it is K -linearly equivalent to the sum of an ample K -divisor and an effective K -divisor.

A *log pair* (X, Δ) is a normal variety X and an effective $\mathbb{R}$ -Weil divisor Δ such that $K_X + \Delta$ is $\mathbb{R}$ -Cartier. We say that a log pair (X, Δ) is *log smooth*, if X is smooth and the support of Δ is a divisor with global normal crossings. A projective morphism $g: Y \rightarrow X$ is a *log resolution* of the pair (X, Δ) if the inverse image of Δ and the exceptional locus is a divisor with global normal crossings. We may write

$$K_Y + \Gamma = g^*(K_X + \Delta) + E,$$

where Γ and E are effective, with no common components, $g_*\Gamma = \Delta$ and E is g -exceptional. Note that this decomposition is unique. The *log discrepancy* of a divisor F

$$a(X, \Delta, F) = 1 + \text{mult}_F(E - \Gamma).$$

A *log canonical place* is any valuation whose exceptional divisor has log discrepancy at most zero and the centre of a log canonical place is called a *log canonical centre*. The pair (X, Δ) is *kawamata log terminal* if

there are no log canonical centres. We say that the pair (X, Δ) is *purely log terminal* (resp. *terminal*) if the log discrepancy of any exceptional divisor is greater than zero (resp. greater than one). We say that the pair is *divisorial log terminal* if there is a log resolution $g: Y \rightarrow X$ such that all exceptional divisors $E \subset Y$ have log discrepancy greater than zero.

Given an integral Weil divisor D , we let

$$R(X, D) = \bigoplus_{m \in \mathbb{N}} H^0(X, \mathcal{O}_X(mD)).$$

If T is a subvariety of X , not contained in the base locus of $|D|$, then $|D|_T$ denotes the image of the linear system $|D|$ under restriction to T .

3. PRELIMINARY RESULTS

In this section we will recall several results about finitely generated algebras and in particular we will give a proof of Shokurov's result that the pl-flip exists if and only if the restricted algebra is finitely generated.

Definition 3.1. *Let X be a normal variety, S be a prime divisor and B an integral Weil divisor, whose support does not contain S . The **restricted algebra** $R_S(X, B)$ is the image of the homomorphism $R(X, B) \rightarrow R(S, B|_S)$.*

Theorem 3.2. *Let $f: X \rightarrow Z$ be a pl-flipping contraction with respect to (X, Δ) . Pick an integer k such that $k(K_X + \Delta)$ is integral.*

Then

- (1) *the flip of f exists if and only if the flip of f exists locally over Z .*
- (2) *If $Z = \text{Spec}(A)$ is affine then the flip $f^+: X^+ \rightarrow Z$ exists if and only if the restricted algebra $R_S(X, k(K_X + \Delta))$ is a finitely generated A -algebra.*

We start with the following well known result:

Lemma 3.3. *Let R be a graded algebra which is an integral domain and let d be a positive integer.*

Then R is finitely generated if and only if $R_{(d)}$ is finitely generated.

Proof. Suppose that R is finitely generated. It is easy to write down an action of the cyclic group $\mathbb{Z}_d$ on R so that the invariant ring is $R_{(d)}$. Thus $R_{(d)}$ is finitely generated by a Theorem of E. Noether which states that the ring of invariants of a finitely generated ring under the action of a finite group is finitely generated.

Suppose now that $R_{(d)}$ is finitely generated. Let $f \in R_i$. Then f is a root of the monic polynomial $x^d - f^d \in R_{(d)}[x]$. It follows that R is integral over $R_{(d)}$ and the result follows by another Theorem of E. Noether on finiteness of the integral closure. $\square$

Lemma 3.4. *Let S be a normal prime divisor on X and let B an integral Weil divisor whose support does not contain S .*

- *If $R(X, B)$ is finitely generated then $R_S(X, B)$ is finitely generated.*
- *If $S \sim B$ and $R_S(X, B)$ is finitely generated then $R(X, B)$ is finitely generated.*

Proof. Since there is a surjective homomorphism $\phi : R(X, B) \rightarrow R_S(X, B)$, it is clear that if $R(X, B)$ is finitely generated then $R_S(X, B)$ is finitely generated.

Suppose now that $R_S(X, B)$ is finitely generated and $S \sim B$. Then there is a rational function g_1 such that $(g_1) = B - S$. If we consider the elements of $R(X, B)_m$ as rational functions, then a rational function g belongs to $R(X, B)_m$ if and only if $(g) + mB \geq 0$. But if g is in the kernel of ϕ , then there is an effective divisor S' such that $(g) + mB = S + S'$. It follows that $(g/g_1) + (m-1)B = S'$ so that $g/g_1 = h \in R(X, B)_{m-1}$. But then the kernel of ϕ is the principal ideal generated by g_1 . $\square$

Proof of (3.2). It is well known that the flip $f^+ : X^+ \rightarrow Z$ exists if and only if the sheaf of graded $\mathcal{O}_Z$ -algebras

$$\bigoplus_{m \in \mathbb{N}} f_* \mathcal{O}_X(mk(K_X + \Delta)),$$

is finitely generated, cf. [7, 6.4]. Since this can be checked locally, this gives (1). If $Z = \text{Spec}(A)$ is affine it suffices to check that $R(X, k(K_X + \Delta))$ is a finitely generated A -algebra.

Since Z is affine and $\rho(X/Z) = 1$ it follows that there are real numbers a and b such that

$$a(K_X + \Delta) \sim bS,$$

and as both $-(K_X + \Delta)$ and $-S$ are ample $\mathbb{Q}$ -divisors, we may assume that a and b are both positive integers. By (3.3) it follows that $R(X, k(K_X + \Delta))$ is finitely generated if and only if $R(X, S)$ is finitely generated. Since Z is affine and f is small, S is mobile so that $S \sim S'$ where S' is an effective divisor whose support does not contain S . By (3.4), $R(X, S)$ is finitely generated if and only if $R_S(X, S')$ is finitely generated. Since $a(K_X + \Delta)|_S \sim bS'|_S$ the result follows by (3.3). $\square$

Lemma 3.5. *Let $\pi: X \rightarrow Z$ be a projective morphism to a normal affine variety Z . Let $(X, \Delta = S + B)$ be a log canonical pair, where $S \geq 0$ and $B \geq 0$ are $\mathbb{Q}$ -Cartier divisors such that $\mathbf{B}_+(B)$ does not contain any log canonical centre of (X, Δ) .*

Then we may find a log canonical pair $(X, \Theta = S + C + D)$ where every log canonical place of (X, Θ) is a log canonical place of (X, Δ) ,

$$K_X + \Theta \sim_{\mathbb{Q}} K_X + \Delta,$$

$C \geq 0$ is ample and $D \geq 0$. In particular if $K_X + \Delta$ is purely log terminal or kawamata log terminal then so is $K_X + \Theta$.

Proof. By assumption we may find an ample $\mathbb{Q}$ -divisor $A \geq 0$ and an effective divisor E such that $B \sim_{\mathbb{Q}} A + E$, where E does not contain any log canonical centre of (X, Δ) . Pick l such that lA is very ample and let $lC \in |lA|$ be a general element. Then for any $\epsilon > 0$ sufficiently small

$$K_X + \Theta = K_X + S + \epsilon C + (1 - \epsilon)B + \epsilon E \sim_{\mathbb{Q}} K_X + \Delta,$$

is log canonical, and every log canonical place of (X, Θ) is a log canonical place of (X, Δ) . $\square$

4. MULTIPLIER IDEAL SHEAVES

The main result of this section is

Theorem 4.1. *Let $\pi: X \rightarrow Z$ be a projective morphism to an affine normal variety Z and let D be a Cartier divisor on X and $M \leq D|_S$ a Cartier divisor on S . Suppose that $D \sim_{\mathbb{Q}} k(K_X + \Delta)$, where k is a positive integer, $(X, \Delta = S + A + B)$ is a purely log terminal pair, $S = \lfloor \Delta \rfloor$ is irreducible, $A \geq 0$ is an ample $\mathbb{Q}$ -divisor and B is an effective $\mathbb{Q}$ -divisor. Assume that X and S are smooth.*

Suppose that there is a positive integer m and a Cartier divisor M on S such that

- (1) $C = mD|_S - M \geq 0$,
- (2) $|M + mA|_S + C \subset |mD + mA|_S$,
- (3) $M - mM + mA|_S$ is very ample and
- (4) $B|_S + \frac{1}{m}C - (D|_S - M) \leq \Xi$ where (S, Ξ) is kawamata log terminal.

Then

$$|M| + (D|_S - M) \subset |D|_S.$$

We have the following immediate consequence:

Corollary 4.2. *Let $\pi: X \longrightarrow Z$ be a projective morphism to an affine normal variety Z and let D be a Cartier divisor on X and $M \leq D|_S$ a Cartier divisor on S . Suppose that $D \sim_{\mathbb{Q}} k(K_X + \Delta)$, where k is a positive integer, $(X, \Delta = S + A + B)$ is a purely log terminal pair, $S = \lfloor \Delta \rfloor$ is irreducible, $A \geq 0$ is an ample $\mathbb{Q}$ -divisor and B is an effective $\mathbb{Q}$ -divisor. Assume that X and S are smooth.*

If there are infinitely many positive integers m such that

$$|mM + mA|_S| + m(D|_S - M) \subset |mD + mA|_S,$$

then

$$|M| + (D|_S - M) \subset |D|_S.$$

Proof. Immediate from (4.1), by letting $M_m = mM$. $\square$

(4.2) becomes more powerful the closer A is to being Cartier. For example we have the following (that we will not use in what follows):

Corollary 4.3. *Let $\pi: X \longrightarrow Z$ be a projective morphism to an affine normal variety Z and let N be a Cartier divisor on X . Suppose that $N \sim_{\mathbb{Q}} k(K_X + \Delta)$, where k is a positive integer, $(X, \Delta = S + A + B)$ is purely log terminal, $S = \lfloor \Delta \rfloor$ is irreducible, $A \geq 0$ is ample and $B \geq 0$.*

If X and S are smooth and A is Cartier then the natural restriction map

$$H^0(X, \mathcal{O}_X(mN)) \longrightarrow H^0(S, \mathcal{O}_S(mN)),$$

is surjective for all $m > 0$.

To prove (4.1), we need a variant of multiplier ideal sheaves:

Definition-Lemma 4.4. *Let (X, Δ) be a log smooth pair where Δ is a reduced divisor and let $|V|$ be a linear system whose base locus contains no log canonical centres of (X, Δ) . Let $\mu: Y \longrightarrow X$ be a log resolution of $|V| + \Delta$ and let F be the fixed divisor of the linear system $\mu^*|V|$. Let $K_Y + \Gamma = \mu^*(K_X + \Delta) + E$ where $\Gamma = \sum P_i$ is the sum of the divisors on Y such that $a(P_i, X, \Delta) = 0$.*

*Then for any real number $c > 0$, define the **multiplier ideal sheaf***

$$\mathcal{J}_{\Delta, c \cdot |V|} := \mu_* \mathcal{O}_Y(E - \lfloor cF \rfloor).$$

If $\Delta = 0$ we will write $\mathcal{J}_{c \cdot |V|}$ or $\mathcal{J}(c \cdot |V|)$ and if $D = cF$, where F is a Cartier divisor, we define

$$\mathcal{J}_{\Delta, D} := \mathcal{J}_{\Delta, c \cdot |V|},$$

where $|V| = F$.

Proof. We have to show that the definition of the multiplier ideal sheaf is independent of the choice of log resolution. Let $\mu: Y \rightarrow X$ and $\mu': Y' \rightarrow X$ be two log resolutions of (X, Δ) and $|V|$. We may assume that μ' factors through μ via a morphism $\nu: Y' \rightarrow Y$. Then $F' = \nu^*F$ as μ^*V is free, and

$$\begin{aligned} E' - cF' &= K_{Y'} + \Gamma' - \mu'^*(K_X + \Delta) - cF' \\ &= K_{Y'} + \Gamma' - \nu^*(K_Y + \Gamma - E + cF) \\ &= \nu^*(E - \lfloor cF \rfloor) + K_{Y'} + \Gamma' - \nu^*(K_Y + \Gamma + \{cF\}) \\ &= \nu^*(E - \lfloor cF \rfloor) + G. \end{aligned}$$

Since $(Y, \text{Supp}(\Gamma + E + F))$ is log smooth, it follows that $(Y, \Gamma + \{cF\})$ is log canonical and has the same log canonical places as (Y, Γ) and hence as (X, Δ) . Thus $\lceil G \rceil \geq 0$ and since $\nu_*(K_{Y'} + \Gamma') = K_Y + \Gamma$, $\lceil G \rceil$ is ν -exceptional. Then

$$\begin{aligned} \mu'_* \mathcal{O}_{Y'}(\lceil E' - cF' \rceil) &= \mu'_*(\nu_* \mathcal{O}_{Y'}(\lceil E' - cF' \rceil)) \\ &= \mu'_*(\nu_* \mathcal{O}_{Y'}(\nu^*(\lceil E - cF \rceil) + \lceil G \rceil)) \\ &= \mu_* \mathcal{O}_Y(\lceil E - cF \rceil). \end{aligned} \quad \square$$

We need to develop a little of the theory of multiplier ideal sheaves.

Lemma 4.5. *Let (X, Δ) be a log smooth pair where Δ is reduced, let $|V|$ be a linear system whose base locus contains no log canonical centres of (X, Δ) and let $G \geq 0$ and $D \geq 0$ be $\mathbb{Q}$ -Cartier divisors whose supports contain no log canonical centres of (X, Δ) .*

Then

- (1) $\mathcal{J}_{\Delta, D} = \mathcal{O}_X$ if and only if $(X, \Delta + D)$ is divisorially log terminal and $\lfloor \Delta + D \rfloor = \lfloor \Delta \rfloor$.
- (2) If $0 \leq \Delta' \leq \Delta$, then $\mathcal{J}_{\Delta, c|V|} \subset \mathcal{J}_{\Delta', c|V|}$. In particular, $\mathcal{J}_{\Delta, c|V|} \subset \mathcal{J}_{c|V|} \subset \mathcal{O}_X$.
- (3) If $\Sigma \geq 0$ is a Cartier divisor, $D \leq G + \Sigma$ and $\mathcal{J}_{\Delta, G} = \mathcal{O}_X$ then $\mathcal{I}_\Sigma \subset \mathcal{J}_{\Delta, D}$.

Proof. (1) follows easily from the definitions.

(2) follows from the fact that $a(P, X, \Delta') \geq a(P, X, \Delta)$ for all divisors P on Y .

To see (3), let $\mu: Y \rightarrow X$ be a log resolution of $(X, \text{Supp}(D+G+\Sigma))$ and note that as Σ is integral, we have

$$\lfloor \mu^*D \rfloor \leq \lfloor \mu^*G \rfloor + \mu^*\Sigma,$$

and as $\mathcal{J}_{\Delta, G} = \mathcal{O}_X$, we have

$$E - \lfloor \mu^*G \rfloor \geq 0.$$

Therefore,

$$\begin{aligned}
-\mu^*\Sigma &\leq \lrcorner\mu^*G\lrcorner - \lrcorner\mu^*D\lrcorner \\
&= (\lrcorner\mu^*G\lrcorner - E) + (E - \lrcorner\mu^*D\lrcorner) \\
&\leq E - \lrcorner\mu^*D\lrcorner.
\end{aligned}$$

Hence

$$\mathcal{I}_\Sigma = \mu_*\mathcal{O}_Y(-\mu^*\Sigma) \subset \mu_*\mathcal{O}_Y(E - \lrcorner\mu^*D\lrcorner) = \mathcal{J}_{\Delta,D}. \quad \square$$

We now recall several vanishing results which are important in applications of multiplier ideal sheaves.

Theorem 4.6. (*Nadel Vanishing*) *Let X be a smooth variety and let Δ be an effective $\mathbb{Q}$ -divisor. Let $f: X \rightarrow Z$ be any projective morphism and let N be any Cartier divisor such that $N - \Delta$ is relatively big and nef.*

Then

$$R^i f_*(\mathcal{O}_X(K_X + N) \otimes \mathcal{J}_\Delta) = 0,$$

for $i > 0$.

We will need the following immediate consequence:

Corollary 4.7. *Let X be a smooth variety, let $\Delta = \sum \delta_i \Delta_i$ be an effective $\mathbb{Q}$ -divisor with simple normal crossings support and $0 < \delta_i \leq 1$. Let $f: X \rightarrow Z$ be any projective morphism and let N be any Cartier divisor such that $N - \Delta$ is relatively big and nef and its restriction to any intersection $\Delta_{i_0} \cap \dots \cap \Delta_{i_t}$ of components of $\lrcorner\Delta\lrcorner$ is relatively big.*

Then

$$R^i f_* \mathcal{O}_X(K_X + N) = 0, \quad \text{for } i > 0.$$

Proof. We proceed by induction on the dimension of X . Note that if $\dim X = 1$, then N is ample and the assertion is clear. If $\lrcorner\Delta\lrcorner = 0$, the assertion follows from Theorem 4.6. Assume now that $\lrcorner\Delta\lrcorner \neq 0$ and let S be an irreducible component of $\lrcorner\Delta\lrcorner$. Then S is a smooth divisor and we have a short exact sequence

$$0 \rightarrow \mathcal{O}_X(K_X + N - S) \rightarrow \mathcal{O}_X(K_X + N) \rightarrow \mathcal{O}_S(K_S + (N - S)|_S) \rightarrow 0.$$

By induction on the number of components of $\lrcorner\Delta\lrcorner$,

$$R^i f_* \mathcal{O}_X(K_X + N - S) = 0,$$

for all $i > 0$ and by induction on the dimension of X ,

$$R^i f_* \mathcal{O}_S(K_S + (N - S)|_S) = 0,$$

for all $i > 0$. $\square$

We have the following extension of (9.5.1) of [8] or (2.4.2) of [12]:

Lemma 4.8. *Let (X, Δ) be a log smooth pair where Δ is reduced, let N be a Cartier divisor, let S be a component of Δ , let $D \geq 0$ be a $\mathbb{Q}$ -Cartier which does not contain any log canonical centre of (X, Δ) and let $\Theta = (\Delta - S)|_S$.*

(1) *There is a short exact sequence*

$$0 \longrightarrow \mathcal{J}_{\Delta-S, D+S}(N) \longrightarrow \mathcal{J}_{\Delta, D}(N) \longrightarrow \mathcal{J}_{\Theta, D|_S}(N) \longrightarrow 0.$$

(2) *If $N - D - (K_X + \Delta)$ is nef and big, and its restriction to any log canonical centre of (X, Δ) is big, then the map*

$$H^0(X, \mathcal{J}_{\Delta, D}(N)) \longrightarrow H^0(S, \mathcal{J}_{\Theta, D|_S}(N)),$$

is surjective.

Proof. By the resolution lemma of [11], we may find a log resolution $\mu: Y \longrightarrow X$ of $(X, \Delta + D)$ which is an isomorphism over the generic point of each log canonical centre of (X, Δ) . Let T be the strict transform of S . If we set $B = \mu^*N + E - \mu^*D$ then we have a short exact sequence

$$0 \longrightarrow \mathcal{O}_Y(\ulcorner B^\nabla - T) \longrightarrow \mathcal{O}_Y(\ulcorner B^\nabla) \longrightarrow \mathcal{O}_T(\ulcorner B^\nabla) \longrightarrow 0.$$

Now $\mu_*\mathcal{O}_Y(\ulcorner B^\nabla) = \mathcal{J}_{\Delta, D}(N)$. If Γ is the sum of the divisors of log discrepancy zero then

$$E - \mu^*D = (K_Y + \Gamma) - \mu^*(K_X + \Delta + D).$$

But then

$$E - \mu^*D - T = (K_Y + \Gamma - T) - \mu^*(K_X + \Delta - S + (D + S)),$$

so that

$$\mu_*\mathcal{O}_Y(\ulcorner B^\nabla - T) = \mathcal{J}_{\Delta-S, D+S}(N),$$

and

$$(E - \mu^*D)|_T = K_T + (\Gamma - T)|_T - \mu^*(K_S + \Theta + D|_S),$$

so that

$$\mu_*\mathcal{O}_T(\ulcorner B^\nabla) = \mathcal{J}_{\Theta, D|_S}(N).$$

On the other hand,

$$\ulcorner B^\nabla - T - (K_Y + \Gamma - T + \{\mu^*D\}) = \mu^*(N - K_X - \Delta - D),$$

is μ -nef and μ -big and its restriction to any log canonical centre of $(Y, \Gamma - T + \{\mu^*D\})$ is μ -big so that by (4.7)

$$R^1\mu_*\mathcal{O}_Y(\ulcorner B^\nabla - T) = 0.$$

This proves (1).

To see (2), note that by (4.7), we have

$$H^1(Y, \mathcal{O}_Y(\ulcorner B^\nabla - T)) = 0,$$

and since

$$R^1\mu_*\mathcal{O}_Y(\ulcorner B^\top - T) = 0,$$

it follows that

$$H^1(X, \mathcal{J}_{\Delta-S, D+S}(N)) = H^1(Y, \mu_*\mathcal{O}_Y(\ulcorner B^\top - T)) = 0.$$

This proves (2). $\square$

Proof of (4.1). Let $\Sigma \in |M|$. Pick a general divisor $H \in |M - mM + mA|_S$. Since

$$M + mA|_S = mM + M - mM + mA|_S,$$

we may pick a divisor

$$G \in |mD + mA| \quad \text{such that} \quad G|_S = m\Sigma + H + C.$$

Let

$$\Lambda = \frac{k-1}{mk}G + B \quad \text{and} \quad N = D - K_X - S.$$

Then Λ is effective, S is not contained in its support and

$$N - \Lambda \sim_{\mathbb{Q}} \frac{1}{k}A,$$

is nef and big. By (4.8), sections of $H^0(S, \mathcal{J}_{\Lambda|_S}(D))$ extend to sections of $H^0(X, \mathcal{O}_X(D))$. Since H is a general very ample divisor, and (S, Ξ) is kawamata log terminal, it follows that $(S, \Xi + \frac{1}{m}H)$ is kawamata log terminal for any $m > 1$. But

$$\begin{aligned} \Lambda|_S - (\Sigma + D|_S - M) &= \frac{k-1}{mk}(m\Sigma + H + C) + B - (\Sigma + D|_S - M) \\ &\leq \frac{1}{m}(H + C) + B - (D|_S - M) \\ &\leq \Xi + \frac{1}{m}H, \end{aligned}$$

and we are done by (3) of (4.5). $\square$

Proof of (4.3). Consider the statement $S(i)$ that for all $m \in n\mathbb{N}$ the map

$$H^0(X, \mathcal{O}_X(m(N + iA))) \longrightarrow H^0(S, \mathcal{O}_S(m(N + iA))),$$

is surjective.

If $N + iA$ is ample then

$$m(N + iA) - S \sim_{\mathbb{Q}} K_X + B + C,$$

where

$$C = (mk-1)(K_X + S + A + B) + (mi+1)A \sim_{\mathbb{Q}} \frac{mk-1}{k}(N + iA) + \left(\frac{i}{k} + 1\right)A,$$

is ample. In particular $H^1(X, \mathcal{O}_X(m(N + iA) - S)) = 0$ by Kawamata-Viehweg vanishing. Considering the natural restriction exact sequence, it follows that $S(i)$ holds for all i sufficiently large. On the other hand, (4.2) proves that $S(i + 1)$ implies $S(i)$. $\square$

5. ASYMPTOTIC MULTIPLIER IDEAL SHEAVES

Definition 5.1. *Let X be a normal variety and let D be a divisor. An **additive sequence of linear systems associated to D** is a sequence $V_\bullet$, such that $V_m \subset H^0(X, \mathcal{O}_X(mD))$ and*

$$|V_i| + |V_j| \subset |V_{i+j}|.$$

Definition-Lemma 5.2. *Suppose that (X, Δ) is log smooth and Δ is reduced and let $V_\bullet$ be an additive sequence of linear systems associated to a divisor D . Assume that there is an integer $m > 0$ such that no log canonical center of (X, Δ) is contained in $\text{Bs}(|V_m|)$.*

If c is a positive real number and p and q are positive integers divisible by m then

$$\mathcal{J}_{\Delta, \frac{c}{p} \cdot |V_p|} \subset \mathcal{J}_{\Delta, \frac{c}{q} \cdot |V_q|} \quad \forall q \text{ divisible by } p.$$

*In particular the **asymptotic multiplier ideal sheaf of $V_\bullet$***

$$\mathcal{J}_{\Delta, c \cdot V_\bullet} = \bigcup_{p > 0} \mathcal{J}_{\Delta, \frac{c}{p} \cdot |V_p|},$$

is given by $\mathcal{J}_{\Delta, c \cdot V_\bullet} = \mathcal{J}_{\Delta, \frac{c}{p} \cdot |V_p|}$, for p sufficiently large and divisible. If we take $V_m = |mD|$ the complete linear system, then define

$$\mathcal{J}_{\Delta, c \cdot \|D\|} = \mathcal{J}_{\Delta, c \cdot V_\bullet},$$

and if S is a component of Δ and we take $W_m = |mD|_S$, then define

$$\mathcal{J}_{\Theta, c \cdot \|D\|_S} = \mathcal{J}_{\Theta, c \cdot W_\bullet},$$

where $\Theta = (\Delta - S)|_S$.

Proof. If p divides q then pick a common log resolution $\mu: Y \rightarrow X$ of V_p and V_q and note that

$$\frac{1}{q} F_q \leq \frac{1}{p} F_p,$$

where F_p is the fixed locus of $\mu^*|V_p|$ and F_q is the fixed locus of $\mu^*|V_q|$. Therefore $\mathcal{J}_{\Delta, \frac{c}{p} \cdot |V_p|} \subset \mathcal{J}_{\Delta, \frac{c}{q} \cdot |V_q|}$. The equality $\mathcal{J}_{\Delta, c \cdot V_\bullet} = \mathcal{J}_{\Delta, \frac{c}{p} \cdot |V_p|}$, now follows as X is Noetherian. $\square$

We are now ready to state the main result of this section:

Theorem 5.3. *Let $\pi: X \rightarrow Z$ be a projective morphism to a normal affine variety Z . Suppose that $(X, \Delta = S + B)$ is log smooth and purely log terminal, where $S = \lfloor \Delta \rfloor$ is irreducible and let k be a positive integer such that $D = k(K_X + \Delta)$ is integral. Let A be any ample divisor on X and let $P = pA$, where p is sufficiently large and divisible.*

If the stable base locus of D does not contain any log canonical centre of $(X, \lceil \Delta \rceil)$, then

$$\mathcal{J}_{\|mD\|_S} \subset \mathcal{J}_{\Theta, \|mD+P\|_S} \quad \text{for all } m \in \mathbb{N},$$

where $\Theta = \lceil B \rceil|_S$.

We will need some results about the sheaves $\mathcal{J}_{\Delta, c \cdot V_\bullet}$, most of which are easy generalizations of the corresponding facts for the usual asymptotic multiplier ideal sheaves.

Lemma 5.4. *Let $\pi: X \rightarrow Z$ be a projective morphism to a normal affine variety Z and let D be a $\mathbb{Q}$ -Cartier divisor. Suppose that (X, Δ) is log smooth, Δ is reduced and the stable base locus of D contains no log canonical center of (X, Δ) . Then*

- (1) *for any real numbers $0 < c_1 \leq c_2$ there is a natural inclusion*

$$\mathcal{J}_{\Delta, c_2 \cdot \|D\|} \subset \mathcal{J}_{\Delta, c_1 \cdot \|D\|},$$

- (2) *if D is Cartier and S is a component of Δ , then the image of the map*

$$\pi_* \mathcal{O}_X(D) \rightarrow \pi_* \mathcal{O}_S(D),$$

is contained in $\pi_ \mathcal{J}_{\Theta, \|D\|_S}(D)$ where $\Theta = (\Delta - S)|_S$.*

Proof. (1) is immediate from the definitions.

Suppose that D is Cartier. Pick an integer p such that

$$\mathcal{J}_{\Theta, \|D\|_S} = \mathcal{J}_{\Theta, \frac{1}{p} \cdot |pD|_S},$$

and a log resolution $\mu: Y \rightarrow X$ of $|D|$ and $|pD|$. Let F_1 be the fixed locus of $\mu^*|D|$ and let F_p be the fixed locus of $\mu^*|pD|$. We have

$$(\pi \circ \mu)_* \mathcal{O}_Y(\mu^*D - F_1) = \pi_* \mathcal{O}_X(D) = (\pi \circ \mu)_* \mathcal{O}_Y(E + \mu^*D).$$

The first equality follows by definition of F_1 and the second follows as E is effective and exceptional. As there are inequalities

$$\mu^*D - F_1 \leq \mu^*D - \lfloor F_p/p \rfloor \leq E + \mu^*D - \lfloor F_p/p \rfloor \leq E + \mu^*D.$$

the image of $\pi_* \mathcal{O}_X(D)$ is equal to the image of

$$(\pi \circ \mu)_* \mathcal{O}_Y(E + \mu^*D - \lfloor F_p/p \rfloor).$$

Thus the image of $\pi_* \mathcal{O}_X(D)$ is contained in

$$(\pi \circ \mu)_* \mathcal{O}_Y(E + \mu^*D - \lfloor F_p/p \rfloor) = \pi_* \mathcal{J}_{\Theta, \|D\|_S}(D). \quad \square$$

Proposition 5.5. *Let $\pi: X \rightarrow Z$ be a projective morphism to a normal affine variety Z and let D be a Cartier divisor. Suppose that (X, Δ) is log smooth and Δ is reduced. Let S be a component of Δ and $\Theta = (\Delta - S)|_S$. If we may find a $\mathbb{Q}$ -Cartier divisor N such that $D - N = A$ is ample, where the stable base locus of N contains no log canonical centres of (X, Δ) , then the image of the map*

$$\pi_* \mathcal{O}_X(K_X + \Delta + D) \rightarrow \pi_* \mathcal{O}_S(K_S + \Theta + D),$$

contains

$$\pi_* \mathcal{J}_{\Theta, \|D\|_S}(K_S + \Theta + D).$$

Proof. Pick an integer p such that

$$\mathcal{J}_{\Theta, \|D\|_S} = \mathcal{J}_{\Theta, \frac{1}{p} \cdot |pD|_S},$$

and a log resolution $\mu: Y \rightarrow X$ of $|pD| + \Delta$ which is an isomorphism over every log canonical centre of (X, Δ) . Let F_p be the fixed divisor of $\mu^*|pD|$, $M_p = p\mu^*D - F_p$ and let Γ and T be the strict transforms of Δ and S . We have a short exact sequence

$$0 \rightarrow \mathcal{O}_Y(G - T) \rightarrow \mathcal{O}_Y(G) \rightarrow \mathcal{O}_T(G) \rightarrow 0,$$

where $G = K_Y + \Gamma + \mu^*D - \lrcorner F_p/p \lrcorner$.

Note that $\lrcorner \Gamma + \{F_p/p\} \lrcorner = \Gamma$ so that $(Y, \Gamma + \{F_p/p\})$ is divisorially log terminal and it has the same log canonical centres as (Y, Γ) . M_p/p is nef, as $|M_p|$ is free. By assumption, for $p \gg 0$, M_p is big and its restriction to every log canonical centre of (Y, Γ) is big. Since μ is an isomorphism over every log canonical centre of (X, Δ) and $G - T \sim_{\mathbb{Q}} K_Y + (\Gamma - T) + \{\frac{1}{p}F_p\} + \frac{1}{p}M_p$, (4.7) implies that

$$R^1 \phi_* \mathcal{O}_Y(G - T) = 0,$$

where $\phi = \pi \circ \mu$. But then there is a surjective homomorphism

$$\pi_* \mathcal{O}_X(K_X + \Delta + D) \supset \phi_* \mathcal{O}_Y(G) \rightarrow \phi_* \mathcal{O}_T(G) = \pi_* \mathcal{J}_{\Theta, \|D\|_S}(K_S + \Theta + D). \quad \square$$

Proof of (5.3). We follow the argument of [4] which in turn is based on the ideas of [6], [10] and [13].

We proceed by induction on m . The statement is clear for $m = 0$, and so it suffices to show that

$$\mathcal{J}_{\|(m+1)D\|_S} \subset \mathcal{J}_{\Theta, \|(m+1)D+P\|_S},$$

assuming that

$$\mathcal{J}_{\|tD\|_S} \subset \mathcal{J}_{\Theta, \|tD+P\|_S} \quad \text{for all} \quad t \leq m.$$

Let $\Delta - S = \sum \delta_i \Delta_i$. For any $1 \leq s \leq k$, put

$$\Xi^s = \sum_{i|\delta_i > (k-s)/k} \Delta_i.$$

We have

- each Ξ^s is integral,
- $0 \leq \Xi^1 \leq \Xi^2 \leq \dots \leq \Xi^k = \lceil \Delta - S \rceil$, and
- $\Delta - S = \frac{1}{k} \sum_{s=1}^k \Xi^s$,

and these properties uniquely determine the divisors Ξ^s . We recursively define integral divisors $D_{\leq s}$ by the rule

$$D_{\leq s} = \begin{cases} 0 & \text{if } s = 0 \\ K_X + S + \Xi^s + D_{\leq s-1} & 1 \leq s \leq k. \end{cases}$$

Note that $D_{\leq k} = D$. Since p is sufficiently large, we may assume that the divisors $\bar{D}_{\leq s} + P$ are sufficiently ample for any $1 \leq s \leq k$. As there is an inclusion

$$\mathcal{J}_{|(m+1)D|_S} \subset \mathcal{J}_{|mD|_S},$$

(cf. (1) of (5.4)), it suffices to prove that there are inclusions

$$(\star) \quad \mathcal{J}_{|mD|_S} \subset \mathcal{J}_{\Theta^{s+1}, |mD + D_{\leq s} + P|_S},$$

for $0 \leq s \leq k$, where $\Theta^i = \Xi^i|_S$ and $\Theta^{k+1} = \Theta^k$.

We proceed by induction on s . Now

$$\mathcal{J}_{|mD|_S} \subset \mathcal{J}_{\Theta, |mD + P|_S} \subset \mathcal{J}_{\Theta^1, |mD + P|_S}.$$

The first inclusion holds by assumption and since $\Theta^1 \leq \Theta$, (2) of (4.5) implies the second inclusion. Thus $(\star)$ holds when $s = 0$.

Now suppose that $(\star)$ holds for $s \leq t-1$. Then

$$\begin{aligned} \pi_* \mathcal{J}_{|mD|_S}(mD + D_{\leq t} + P) &\subset \pi_* \mathcal{J}_{\Theta^t, |mD + D_{\leq t-1} + P|_S}(mD + D_{\leq t} + P) \\ &\subset \text{Im}(\pi_* \mathcal{O}_X(mD + D_{\leq t} + P) \longrightarrow \pi_* \mathcal{O}_S(mD + D_{\leq t} + P)) \\ &\subset \pi_* \mathcal{J}_{\Theta^{t+1}, |mD + D_{\leq t} + P|_S}(mD + D_{\leq t} + P). \end{aligned}$$

The first inclusion holds as we are assuming $(\star)$ for $s = t-1$, the second inclusion holds by (5.5) and the identity

$$mD + D_{\leq t} + P = K_X + S + \Xi^t + (D_{\leq t-1} + P) + mD,$$

and the last inclusion follows from (2) of (5.4). Arguing as in [6], one sees that

$$\mathcal{J}_{|mD|_S}(mD + D_{\leq t} + P),$$

is generated by global sections and so

$$\mathcal{J}_{\|mD\|_S} \subset \mathcal{J}_{\Theta^{t+1}, \|mD+D_{\leq t}+P\|_S}. \quad \square$$

6. LIFTING SECTIONS

Theorem 6.1. *Let $\pi: X \rightarrow Z$ be a projective morphism to a normal affine variety Z and let D be an integral divisor on X . Suppose that $D \sim_{\mathbb{Q}} k(K_X + \Delta)$ for some positive integer k , where $(X, \Delta = S + B)$ is a log smooth purely log terminal pair, $S = \lrcorner \Delta \lrcorner$ is irreducible, $B \geq 0$ and $\mathbf{B}_+(B/Z)$ does not contain S .*

If the stable base locus of D contains no log canonical centres of $(X, \lrcorner \Delta \lrcorner)$ then the map

$$H^0(X, \mathcal{O}_X(D)) \rightarrow H^0(S, \mathcal{O}_S(D)),$$

is surjective.

Proof. By assumption, we may write $B \sim_{\mathbb{Q}} A + E$ where A is ample and E is an effective divisor whose support does not contain S . By (3.5), we may assume that $\Delta = S + A + B$ where $S = \lrcorner \Delta \lrcorner$, A is ample, $B \geq 0$ and (X, Δ) is a purely log terminal pair.

Pick $\tau \in H^0(S, \mathcal{O}_S(mD))$ and $\sigma \in H^0(X, \mathcal{O}_X(mA))$. Then

$$\sigma \cdot \tau \in H^0(S, \mathcal{J}_{\|mD\|_S}((mD + mA)|_S)).$$

But it is shown in the proof of (5.3), that for $m \gg 0$, $H^0(X, \mathcal{J}_{\|mD\|_S}(mD + mA))$ is contained in the image of

$$H^0(X, \mathcal{O}_X(mD + mA)) \rightarrow H^0(S, \mathcal{O}_S(mD + mA)).$$

The result follows from (4.2). $\square$

7. RATIONALITY OF THE RESTRICTED ALGEBRA

In this section we will prove:

Theorem 7.1. *Let $\pi: X \rightarrow Z$ be a projective morphism to an affine normal variety Z and let D be an integral divisor. Suppose that $D \sim_{\mathbb{Q}} k(K_X + \Delta)$ for some positive integer k , where $(X, \Delta = S + A + B)$ is a log smooth purely log terminal pair, $S = \lrcorner \Delta \lrcorner$ is irreducible, $A \geq 0$ is a general ample $\mathbb{Q}$ -divisor and B is an effective $\mathbb{Q}$ -divisor. Let $\Omega = (A + B)|_S$. Assume that (S, Ω) is canonical and S is not a component of the base locus of $|D|$. Let $F_m = \text{Fix}(|mD|_S)$, let*

$$F = \liminf \frac{F_m}{mk},$$

and let $\Theta = \Omega - \Omega \wedge F$.

Then Θ is rational and if moreover l is sufficiently divisible, then $lk(K_S + \Theta)$ is Cartier and

$$|lk(K_S + \Theta)| + lk(\Omega - \Theta) = |lD|_S.$$

In particular

$$R_S(X, lD) \cong R(S, lk(K_S + \Theta)).$$

We will need the following.

Theorem 7.2. *Let $\pi: X \rightarrow Z$ be a projective morphism to an affine normal variety Z and let D be an integral divisor. Suppose that $D \sim_{\mathbb{Q}} k(K_X + \Delta)$ for some positive integer k , where $(X, \Delta = S + A + B)$ is a log smooth purely log terminal pair, $S = \lfloor \Delta \rfloor$ is irreducible, $A \geq 0$ is a general ample $\mathbb{Q}$ -divisor and B is an effective $\mathbb{Q}$ -divisor. Let $\Omega = (A + B)|_S$. Assume that (S, Ω) is canonical and S is not a component of the base locus of $|D|$. Let m be a positive integer and*

$$\Theta_m = \Omega - \Omega \wedge \frac{F_m}{mk}.$$

Then $mk\Theta_m$ is integral and

(1) *If H is a semiample divisor on X , then*

$$|mk(K_S + \Theta_m) + H|_S| + mk(\Omega - \Theta_m) = |mD + H|_S.$$

(2) *If $l > 0$ is a fixed integer, Φ is a $\mathbb{Q}$ -divisor on S such that $\Phi \leq \Omega$ and $M = lk(K_S + \Phi)$ is integral and there are infinitely many integers $m > 0$ such that $mlk(\Theta_{ml} - \Phi) + mA|_S$ is very ample and $(S, \Omega + lk(\Phi - \Theta_{ml}))$ is kawamata log terminal, then*

$$|M| + lk(\Omega - \Phi) \subset |lD|_S.$$

Proof. Let $g: Y \rightarrow X$ be a log resolution of $|mk(K_X + \Delta)| + \Delta$. We may write

$$K_Y + \Psi = g^*(K_X + \Delta) + E$$

where $\Psi \geq 0$ and $E \geq 0$ have no common components, $g_*\Psi = \Delta$ and $g_*E = 0$. We may assume that the components of $\Psi - (g^{-1})_*S$ are disjoint. Let $C = g^*A$. Note that $C = (g^{-1})_*A$ so that $C \leq \Psi$. Let

$$\Gamma = \Psi - \Psi \wedge \frac{\text{Fix}|mk(K_Y + \Psi)|}{mk}.$$

Pick a $\mathbb{Q}$ -divisor G such that pG is a general element of $|pg^*H|$ for some $p \gg 0$ sufficiently divisible. By (7.3) one sees that $mk\Theta_m$ is integral and no log canonical center of $(Y, \lceil \Gamma + G \rceil)$ is contained in the base locus of $mk(K_Y + \Gamma + \frac{1}{mk}G)$. Let $T = (g^{-1})_*S$ and $h = g|_T: T \rightarrow S$. Since $\Gamma \geq C$ and there is an effective and exceptional divisor F such that $C - F$ is ample, it follows that $\mathbf{B}_+(\Gamma - T + \frac{1}{mk}G/Z)$ does not

contain T . By (6.1) (applied to $D = mk(K_Y + \Gamma + \frac{1}{mk}g^*H)$) we have that

$$H^0(Y, \mathcal{O}_Y(mk(K_Y + \Gamma) + g^*H)) \rightarrow H^0(T, \mathcal{O}_T(mk(K_Y + \Gamma) + g^*H))$$

is surjective. Therefore

$$|(mk(K_Y + \Gamma) + g^*H)|_T| + mk(\Psi - \Gamma)|_T \subset |mk(K_Y + \Psi) + g^*H|_T.$$

By (7.3), $h_*(K_Y + \Gamma)|_T = K_S + \Theta_m$ and $mk\Theta_m$ is integral so that pushing forward to X , (1) follows.

(2) follows from (4.1), where the divisors M_m , C_m , M and D correspond to the divisors $mlk(K_S + \Theta_m)$, $mlk(\Omega - \Theta_m)$, M and lD respectively. Condition (1) of (4.1) is immediate and condition (2) follows from what we have just proved. Conditions (3) and (4), follow since we have assumed that $mlk(\Theta_m - \Phi) + mA|_S$ is very ample and $(S, B|_S + lk(\Phi - \Theta_m))$ is kawamata log terminal. $\square$

Lemma 7.3. *Let $(X, \Delta = S + A + B)$ be a log smooth purely log terminal pair such that $S = \lfloor \Delta \rfloor$ is irreducible, $A \geq 0$ is a general ample $\mathbb{Q}$ -divisor and B is an effective $\mathbb{Q}$ -divisor. Assume that $D \sim_{\mathbb{Q}} k(K_X + \Delta)$ for some positive integer $k > 0$. For any $m > 0$, we let $\Theta_m = \Omega - \Omega \wedge \frac{1}{mk}F_m$ where $\Omega = (\Delta - S)|_S$ and $F_m = \text{Fix}(|mD|_S)$. Pick any log resolution $g : Y \rightarrow X$ of $|mD| + \Delta$ and write*

$$K_Y + \Psi = g^*(K_X + \Delta) + E$$

where $\Psi \geq 0$ and $E \geq 0$ have no common components, $g_*\Psi = \Delta$ and $g_*E = 0$. Let

$$\Gamma = \Psi - \Psi \wedge \frac{\text{Fix}(mk(K_Y + \Psi))}{mk}$$

and $h = g|_T : T \rightarrow S$.

If (S, Ω) is canonical and the components of $(\Gamma - T)|_T$ are disjoint, then the base locus of $|mk(K_Y + \Gamma)|$ contains no canonical centers of $(T, (\Gamma - T)|_T)$, $h_*(\Gamma - T)|_T = \Theta_m$, $mk\Theta_m$ is integral and $H^0(T, \mathcal{O}_T(mk(K_Y + \Gamma))) \cong H^0(S, \mathcal{O}_S(mk(K_S + \Theta_m)))$.

Proof. For any divisor P in T given by a component of $\text{Supp}(\Psi - T)|_T$, we have that $\text{mult}_P(E|_T) = 0$ and if P is not h -exceptional then

$$\text{mult}_P(|mD|_S) = \text{mult}_P(|mg^*D|_T) = \text{mult}_P(|mk(K_Y + \Psi)|_T),$$

and $\text{mult}_P(\Omega) = \text{mult}_P((\Psi - T)|_T)$. It follows that for these divisors $\text{mult}_P((\Gamma - T)|_T) = \text{mult}_P\Theta_m$. As (S, Ω) is canonical,

$$(\Gamma - T)|_T = (h^{-1})_*\Theta_m.$$

Therefore $h_*(\Gamma - T)|_T = \Theta_m$ and $H^0(T, \mathcal{O}_T(mk(K_Y + \Gamma))) \cong H^0(S, \mathcal{O}_S(mk(K_S + \Theta_m)))$. Note also that by definition $mk\Gamma$ is integral and so $mk\Theta_m$ is also integral.

As the only canonical centers of $(T, (\Gamma - T)|_T)$ are the components of $(\Gamma - T)|_T \leq (\Psi - T)|_T$ and $\text{Supp}(\Psi) + \text{Fix}(mk(K_Y + \Psi))$ has normal crossings, it follows that no component of $(\Gamma - T)|_T$ is contained in $\text{Fix}(mk(K_Y + \Gamma)) = \text{Bs}(mk(K_Y + \Gamma))$. $\square$

Proof of (7.1). We let $\Theta_m = \Omega - \Omega \wedge \frac{F_m}{mk}$. For any fixed integer $l > 0$ and any integer $m > 0$, we have $F_{ml} \leq lF_m$ so that

$$F = \liminf \frac{F_m}{mk} \geq \liminf \frac{F_{ml}}{mlk} \geq \liminf \frac{F_m}{mk} = F,$$

and hence $\limsup \Theta_{ml} = \Theta$. Passing to a subsequence $I \subset \mathbb{N}$, we may then assume that

$$\lim_{m \in I} \Theta_{ml} = \Theta.$$

Pick a positive rational number $\epsilon > 0$ such that

- $\frac{1}{2}A|_S + G$ is ample for any $\mathbb{Q}$ -divisor G on S such that $-\epsilon\Theta \leq G \leq \epsilon\Theta$,
- $(S, \Omega + \epsilon\Theta)$ is kawamata log terminal.

Let $\Theta = \sum \theta_i P_i$ and l_0 be an integer such that if $\theta_i \in \mathbb{Q}$, then $l_0 \theta_i \in \mathbb{N}$. Let $l > 0$ be an integer divisible by l_0 and $\Phi = \Phi_l$ be a $\mathbb{Q}$ -divisor such that $M = lk(K_S + \Phi)$ is integral and

$$-\epsilon\Theta \leq M - lk(K_S + \Theta) = lk(\Phi - \Theta) \leq \epsilon\Theta.$$

We have

$$lk(\Phi - \Theta_{ml}) = lk(\Phi - \Theta) + lk(\Theta - \Theta_{ml}).$$

Therefore, for $m \in I$ sufficiently big

- (a) $mlk(\Theta_{ml} - \Phi) + mA|_S$ is very ample and
- (b) $(S, \Omega + lk(\Phi - \Theta_{ml}))$ is kawamata log terminal.

By (2) of (7.2), we have

$$|M| + lk(\Omega - \Phi) \subset |lD|_S.$$

It follows that if $\theta_i \notin \mathbb{Q}$ and $\text{mult}_{P_i}(\Phi - \Theta) > 0$, then $P_i \subset \text{Bs}(lk(K_S + \Phi))$. Let $V \subset \text{Div}_{\mathbb{R}}(S)$ be the vector space spanned by the components of $\Theta - A|_S$. Pick a log terminal model $\nu : S \dashrightarrow S'$ for (S, Θ) and let $\mathcal{P} \subset V + A|_S$ be the open rational polytope containing Θ such that ν is a log terminal model for a kawamata log terminal pair (S, Φ) with $\Phi \in V + A|_S$ if and only if $\Phi \in \mathcal{P}$ cf. [2, Corollary 1.1.5]. By [?], after possibly shrinking $\mathcal{P}$, we may assume that there is a positive integer r

such that if $s(K_S + \Phi)$ is integral for some $\Phi \in \mathcal{P}$ and some positive integer s , then $rs(K_{S'} + \nu_*\Phi)$ is base point free.

Suppose that θ_i is not rational, then by Diophantine approximation, we may find an integer $l = l_i$ and a divisor $\Phi = \Phi_{l_i}$ as above such that

- $\Phi \in \mathcal{P}$,
- $\frac{1}{r}lk(K_S + \Phi)$ is integral,
- $-\epsilon\Theta \leq lk(\Phi - \Theta) \leq \epsilon\Theta$, and
- $\text{mult}_{P_i}(\Phi - \Theta) > 0$.

But then, as $P_i \subset \text{Bs}|lk(K_S + \Phi)|$ and $lk(K_{S'} + \nu_*\Phi)$ is base point free, the divisor P_i is contracted by ν . Therefore ν contracts all irrational components of Θ so that for any l sufficiently divisible and any $t > 0$, $\nu_*\Theta_{tl} = \nu_*\Theta_l$. Moreover, as l is sufficiently divisible, we may assume that $tlk(K_{S'} + \nu_*\Theta_{tl}) = tlk(K_{S'} + \nu_*\Theta_l)$ is Cartier and generated. Then

$$|tlk(K_S + \Theta_{tl})| = |tlk(K_S + \Theta_l)| + tlk(\Theta_{tl} - \Theta_l).$$

By (7.2), we have

$$|tlk(K_S + \Theta_{tl})| + tlk(\Omega - \Theta_{tl}) = |tlD|_S.$$

But then $tlk(\Theta_{tl} - \Theta_l)$ is contained in $tlk\Theta_{tl}$ and in $\text{Fix}(|tlD|_S)$ so that $\Theta_{tl} = \Theta_l$ and hence $\Theta = \Theta_l$ is rational. The Theorem now follows by what we have already proven. $\square$

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